



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Roll Number	2520030013
Branch	CSE
Course Outcome	3

Case Study Topic: Delhi-NCR Delivery Routing using Bidirectional Dijkstra

Work Done Summary:

In this case study, we implemented Bidirectional Dijkstra's algorithm to find the shortest delivery route from Connaught Place (CP) to Greater Noida West (GNW) through an 8-node Delhi-NCR road graph. The graph includes nodes: CP, KAR (Karol Bagh), DWA (Dwarka), GUR (Gurgaon), SAK (Saket), NOI (Noida), GZB (Ghaziabad), and GNW, with 11 weighted edges representing travel times in minutes. Bidirectional Dijkstra runs two simultaneous searches — forward from CP and backward from GNW — and stops when they meet, visiting far fewer nodes than standard Dijkstra.

Implementation Details:

1. Built the 8-node weighted undirected graph with all 11 edges from the Delhi-NCR road network.
2. Ran forward Dijkstra from CP: relaxed KAR (15), SAK (12), DWA (18).
3. Expanded SAK (dist=12): relaxed NOI (28) and DWA (32).
4. Ran backward Dijkstra from GNW: relaxed NOI (13), GUR (25), GZB (30).
5. Detected meeting point at NOI — found in both forward and backward frontiers.
6. Reconstructed shortest path: CP → SAK → NOI → GNW = 12 + 16 + 13 = 41 minutes.

Result: Screenshots / Output

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=== Bidirectional Dijkstra: Delhi-NCR Routing ===
Source: CP | Destination: GNW

Forward distances from CP:
CP -> CP = 0
CP -> KAR = 15
CP -> DWA = 18
CP -> GUR = 42
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CP -> SAK = 12
CP -> NOI = 28
CP -> GZB = 37
CP -> GNW = 41
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Backward distances from GNW:

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GNW -> CP = 41
GNW -> KAR = 56
GNW -> DWA = 44
GNW -> GUR = 25
GNW -> SAK = 29
GNW -> NOI = 13
GNW -> GZB = 30
GNW -> GNW = 0
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Meeting Point : NOI

Shortest Path : CP -> SAK -> NOI -> GNW

Total Distance: 41 minutes

Time Complexity: $O((V + E) \log V)$ per direction

Process finished with exit code 0

The Bidirectional Dijkstra algorithm successfully found the shortest delivery route. The meeting point was NOI (Noida), combining a forward cost of 28 (CP→SAK→NOI) and a backward cost of 13 (GNW→NOI), giving a total of 41 minutes. Compared to Standard Dijkstra which visits all 8 nodes, Bidirectional Dijkstra stopped early after visiting only 5 nodes, proving it is more efficient for large-scale delivery routing systems.

GitHub Link: <https://github.com/jagadabinanditha/DSA-Case-Studies>

Student Signature	Faculty Signature